The interval is contractible

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 384-386 and its proof environment opens at line 388 after a blank line; the pairing is by adjacency and the retained proof range 388-398 is the whole environment.

Editorial note. This source has no label-number dump entry for this label (newlabel thm:contr-interval is absent from the pinned dump), so no counter value is imposed and the counted lemma prints with the wrapper numbering; the original number is not claimed.

Editorial note. No context span and no external reference are needed: the statement and proof use no label from outside the retained range.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: contractibility of the interval (author label thm:contr-interval).

Allowed standard prerequisites (editorial).

- The interval as a higher inductive type. The interval is generated by two points and a path between them, so a map out of it is given by two points and a path between their images.. It is the structure the proof has to respect.
- Dependent action on paths. For a dependent function f a path s in the base gives a dependent path from f of one endpoint to f of the other, which must be supplied to define f on the path constructor.. It is the datum the proof constructs by hand.
- Path inverses are inverses. For any path its inverse concatenated with it is equal to reflexivity.. It supplies the canonical element that closes the construction.

Lemma 0.0.1. *The type I is contractible.*

Proof. We prove that for all $x : I$ we have $x =_I 1_I$. In other words we want a function f of type $\prod_{(x:I)}(x =_I 1_I)$. We begin to define f in the following way:

$$f(0_I) :\equiv \mathsf{seg} \quad : 0_I =_I 1_I,$$
$$f(1_I) :\equiv \mathsf{refl}_{1_I} : 1_I =_I 1_I.$$

It remains to define $\mathsf{apd}_f(\mathsf{seg})$, which must have type $\mathsf{seg} =^{\lambda x.\, x =_I 1_I}_{\mathsf{seg}} \mathsf{refl}_{1_I}$. By definition this type is $\mathsf{seg}_*(\mathsf{seg}) =_{1_I =_I 1_I} \mathsf{refl}_{1_I}$, which in turn is equivalent to $\mathsf{seg}^{-1} \cdot \mathsf{seg} = \mathsf{refl}_{1_I}$. But there is a canonical element of that type, namely the proof that path inverses are in fact inverses. □